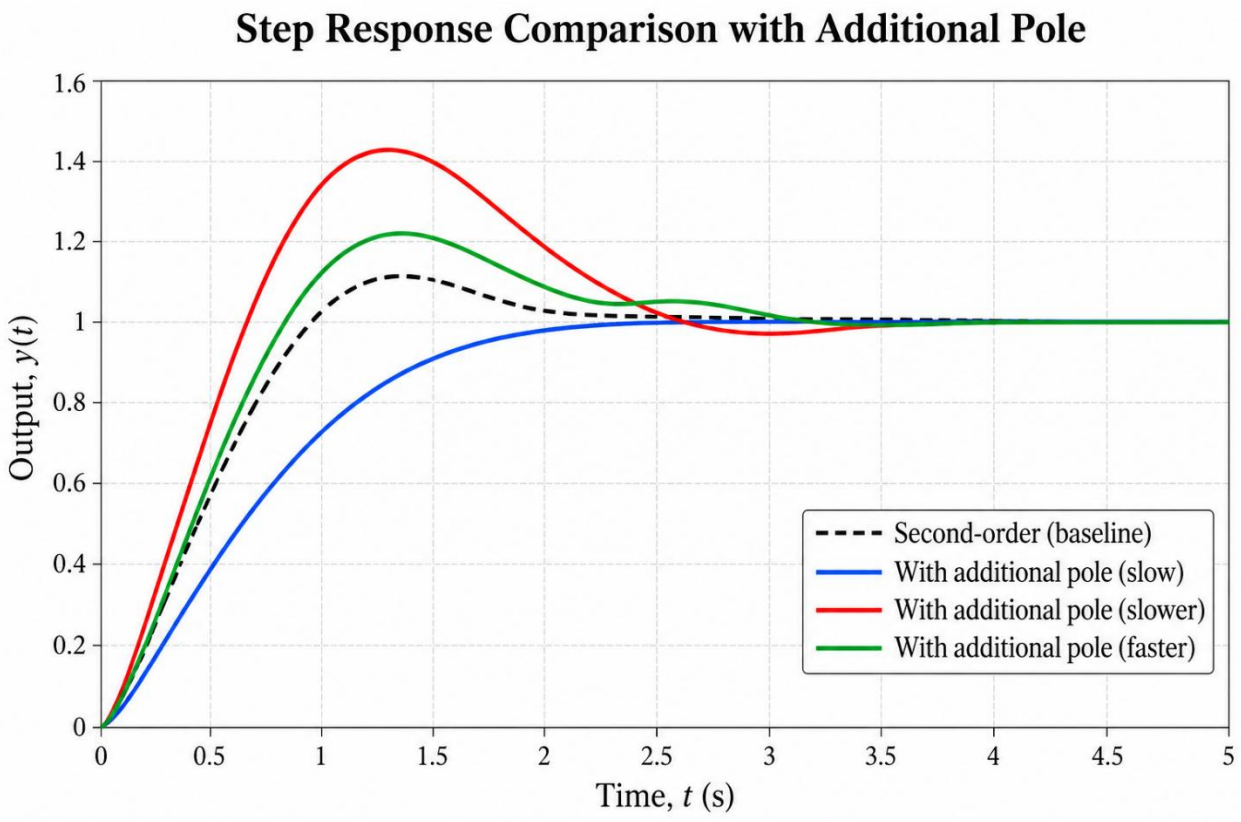
LAB NO: 7

OPEN ENDED LAB

Time Response of Second Order Systems, Additional Pole and zero



Lab Practice Problem 1:

Students who are using MATLAB should now run ch4apB2 in Appendix B. You will learn how to generate a step response for a transfer function and how to plot the response directly or collect the points for future use. The example shows how to collect the points and then use them to create a multiple plot, title the graph, and label the axes and curves to produce
the graph in Figure 4.24 to solve Example 4.8.

Lab Practice Problem 2:

Use the following MATLAB and Control System Toolbox statements to investigate the effect of the additional pole in Skill-Assessment Exercise 4.6(a). Move the higher-order pole originally at −15 to other values by changing "a" in the code.

```
a=15
numga=100*a;
denga=conv([1 a],...[1 4 100]);
Ta=tf (numga,denga);
numg=100;
deng=[1 4 100];
T=tf (numg,deng);
step(Ta,'.',T,'-')
```

Lab Practice Problem 3:

Use the following MATLAB and Symbolic Math Toolbox statements to evaluate the effect of higher-order poles by finding the component parts of the time response of c1(t) and c2(t) in Example 4.10.

```
syms s
C1 = 26.25*(s+4)/…(s*(s + 3.5)*…(s+5)*(s+6));
C2 = 26.25*(s+4)/…(s*(s+4.01)*…(s+5)*(s + 6));
c1 = ilaplace(C1);
'c1'
c1 = vpa(c1,3)
c2 = ilaplace(C2);
c2 = vpa(c2,3)
'c2'
pretty(c2);
```

1. Definition of an Open-Ended Problem

An open-ended engineering problem is a problem for which more than one technically valid solution may exist. Students must select assumptions and parameters, investigate alternatives, evaluate trade-offs, and justify a solution against stated requirements. Marks are awarded for the quality of the reasoning and evidence, not for matching one prescribed design.

In this laboratory, students begin with a common second-order baseline model. They then investigate the effects of an additional pole, an additional zero, and a combined pole-zero modification. Students choose and justify parameter values, compare time-response metrics, and defend their final recommendation in an individual viva and practical modification task.

2. Aim

To investigate how damping ratio and natural frequency affect a second-order system, and to evaluate how additional poles and zeros change transient response, stability, and design trade-offs using analytical calculations and MATLAB simulations.

3. Learning Outcomes

- Model a standard second-order system and identify its poles.
- Predict and measure rise time, peak time, percentage overshoot, settling time, and steady-state value where applicable.
- Modify a transfer function to include an additional pole and/or zero and explain the resulting dynamics.
- Use MATLAB to generate step responses and pole-zero maps and extract performance metrics.
- Compare alternatives and justify a design using quantitative evidence.
- Demonstrate individual understanding by modifying a parameter and explaining the result during a viva.

4. Problem Statement

An engineering team needs a stable control system that follows a unit-step reference command with limited overshoot and a sufficiently fast settling time. Investigate the baseline system and then determine whether additional dynamic elements can improve or degrade its performance. There is no single prescribed final design; students must justify their choices.

4.1 Baseline model

Use the common baseline transfer function:

$$T_2(s) = 16/(s^2 + 4.8s + 16)$$

This corresponds to $\omega_n = 4$ rad/s and $\zeta = 0.6$. The baseline has unity DC gain.

4.2. Design specifications

Requirement	Target / rule
Maximum percentage overshoot	≤ 10%
Settling time (2% criterion)	≤ 2.5 s
Steady-state error to unit step	Zero for the baseline and unity-DC-gain modifications
Stability	All closed-loop poles must be in the open left half-plane
Model comparisons	Baseline, added-pole model, added-zero model, combined model

These are target constraints, not guaranteed outcomes for every modification. Students must determine which models satisfy them. Use the same settling-time convention throughout; MATLAB stepinfo uses a 2% settling threshold by default.

5. Theory and Model Definitions

5.1 Standard second-order model

$$T_2(s) = \omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$$

For an underdamped system ($0 < \zeta < 1$):

$$\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$$
$$T_p = \pi/\omega_d$$
$$M_p(\%) = 100e\text{^}(-\pi\zeta/\sqrt{(1 - \zeta^2)})$$
$$T_s \approx 4/(\zeta\omega_n)$$

These are standard second-order approximations. They should not be applied uncritically to higher-order models with added poles and zeros.

5.2 Additional pole

$T_p(s) = T_2(s) \cdot p/(s + p)$, where $p > 0$. The added factor has a pole at $s = -p$ and unity DC gain.

5.3 Additional zero with a fixed pole

$$T_z(s) = T_2(s) \cdot (s + z)/(s + 20),\ z > 0$$

5.4 Combined modification

$T_pz(s) = T_2(s) \cdot p/(s + p) \cdot (s + z)/(s + 20)$.

The added factors preserve the DC gain of the baseline. They do not guarantee that the transient specifications will be met. The added pole at −20 is part of every zero-modified model and must be included in pole analysis.

6. Required Investigation

Part A — *Second-order baseline*

1. Identify ω_n and ζ for the baseline model and calculate its poles.
2. Predict overshoot and approximate settling time using the second-order formulas.
3. Plot the unit-step response and pole-zero map in MATLAB.
4. Record rise time, peak time, overshoot, and settling time using stepinfo.
5. Investigate at least two alternative second-order designs by changing ω_n and/or ζ. Explain the trade-offs.
6. Select a baseline design for the extension studies and justify the selection.

Part B — *Add an additional pole*

7. Use $T_p(s) = T_2(s)p/(s+p)$.
8. Investigate at least three p values: a relatively slow pole, a pole comparable with the baseline dynamics, and a faster pole.
9. For each model, calculate all poles, plot the response, and record the performance metrics.
10. Explain which modes dominate and whether the extra pole improves or degrades the response.

Part C — *Add an additional zero*

11. Use $T_z(s) = T_2(s)(s+z)/(s+20)$.
12. Investigate at least three z values in the supplied or instructor-approved range.
13. Calculate the poles and zero, compare step responses, and record performance metrics.
14. Explain why the added zero changes the transient response and why the fixed pole at −20 must also be considered.

Part D — *Combined pole and zero*

15. Use $T_pz(s) = T_2(s)[p/(s+p)][(s+z)/(s+20)]$.
16. Investigate at least three combinations of p and z.
17. Compare the combined response with the baseline and the separate modifications.
18. Identify which designs satisfy the stated specifications and recommend one design.